

Sustainability Notes

Sustainability

Defined as meeting our present needs without compromising future generations' ability to meet their own.

Need for Sustainability

- Key to preserving our planet.
- Reduces pollution and conserves resources.
- Creates jobs and stimulates the economy.
- Improves public health.
- Protects biodiversity.
- Protects the natural environment.
- Promotes the choice of non-toxic materials.
- Reduces and reuses resources.
- Minimizes waste.
- Used for life-cycle analysis.

Concept, Approaches, and Significance of Sustainability

To build up sustainability development, the following approaches are proposed:

1. **Developing Appropriate Technology:**

- Locally adaptable, eco-friendly, and resource-efficient.
- Culturally suitable.
- Uses local labor and fewer resources.
- Produces minimum waste.

2. **Reduce, Reuse, Recycle (3-R) Approach:**

- Optimum use of natural resources.
- Using resources repeatedly instead of discarding them.
- Recycling materials into further products.
- Reduces pressure on natural resources.
- Decreases waste generation and pollution.

3. **Providing Environmental Education and Awareness:**

- Changes people's attitudes towards the Earth and environment.

4. **Consumption of Renewable Resources:**

- Consume natural resources without exceeding regeneration capacity.

5. Conservation of Non-Renewable Resources:

- Conserve non-renewable resources by recycling and reusing.

6. Population Control:

- Controlling population growth for sustainability development.

Economic and Social Challenges of Sustainability

1. Economic Sustainability

- Organization's ability to manage resources and generate profits responsibly in the long term.

- **Examples:**

- 1. **Unilever:**

- Strategy to balance sustainability and economic performance.
 - Measures include increasing package recycling, promoting recycled materials, and responsible consumption awareness campaigns.

- 2. **Suez:**

- Reduced emissions related to electricity consumption by 95%. using renewable energy and conservation of natural habitats.

- **Economic Challenges:**

- High rates of unemployment or underemployment.
 - High rates of poverty and low growth.
 - Increasing inequality.
 - Disruption of major economic activities due to pandemics (e.g., tourism).
 - Volatile growth dependent on one source.
 - Low productivity due to poor human capital development.
 - Skills mismatch.
 - Lack of quality jobs.
 - Macroeconomic instability.

2. Social Sustainability

- Strengthening the cohesion and stability of specific social groups.

- **Examples:**

- 1. **CEMEX:**

- Contributes to social development of communities.
 - Offers decent housing through self-building programs and loans with favorable access conditions.

- 2. **Gigante Group:**

- Contributes funds and resources to social causes.
- Provides school materials and grants to improve visual health.
- **Social Challenges:**
 - Difficult to measure but easier to identify.
 - Include human rights, fair labor practices, living conditions, health and safety, wellness, diversity and equity, work-life balance, empowerment, community engagement.

Aspects of Sustainability

Four aspects, with environmental sustainability being fundamental.

1. Environmental Aspect:

- Enhance and maintain biophysical systems.
- Includes natural ecosystems and interactions between them and people.
- Calls for guardianship of the environment.

2. Social Aspect:

- Equity within and between generations, and within and between ethnic and social groups.
- Includes people's mental and physical well-being and community cohesion based on fair distribution of resources.

3. Cultural Aspect:

- Nourish and share attitudes and values that represent diverse world views.
- Political need for all people to express their views freely and participate in decision-making.
- Builds resilience for the future.

4. Economic Aspect:

- Interactions of humans with the natural environment using resources to create goods and services.
- Resource use and waste disposal within the planet's capacity.
- Encourages a fair trading system that equitably distributes benefits and costs.
- Encourages innovation and creativity for a sustainable future.

Relationship Between These Aspects

- Economy is a subset of society, which is entirely dependent on the environment.
- Any impact on the environment affects society and the economy.
- Sustainability issues must be considered holistically and recognize their interdependence.

From Unsustainability to Sustainability

Unsustainability

- Exploiting limited resources for greed instead of needs.
 - **Characteristics:**
 - Forgetting responsibility towards the environment.
 - Degradation of available resources.
 - Not caring about future generations' needs.
 - Unsystematic planning leading to damage.
 - Leads to extreme degradation of the environment and living organisms.
 - **Causes:**
 - Developing countries are responsible for the degradation.
 - Developed countries contribute more to national pollution.
 - Rising population.
 - Treating the environment solely as a resource for humans.
 - Extracting too much material from the lithosphere.
 - Creating problematic chemical compounds.
 - Cutting trees faster than they can grow.
 - Purchasing and using polythene bags.

Characteristics of Sustainability

1. Reduces emission of greenhouse gases to reduce global warming.
2. Uses natural and biodegradable materials.
3. Emphasizes using renewable energy sources.
4. Follows non-polluting construction practices.
5. Protects natural habitats.
6. Improves the quality of human life.
7. Minimizes the depletion of natural resources.
8. Teaches respect and care for all life forms.
9. Ensures future generations can meet their own demands.

Differences Between Sustainability and Unsustainability

S.No.	Sustainability	Unsustainability
1.	Prevention of natural resources.	Exploitation of natural resources is faster than the planet can handle and replenish.
2.	Long-term economic growth.	Not quantifying ecosystem services and increased vulnerability to crises.
3.	Equality, diversity, social cohesion.	Labor laws, human rights, gender inequality, and poor treatment of indigenous people.

Millennium Development Goals (MDGs)

Eight international development goals:

1. Eradicate extreme poverty and hunger.
2. Achieve universal primary education.
3. Promote gender equality and empower women.
4. Reduce child mortality.
5. Improve maternal health.
6. Combat HIV/AIDS, malaria, and other diseases.
7. Ensure environmental sustainability.
8. Develop a global partnership for development.

Sustainability Protocols

Sustainability standards and certifications used by producers, manufacturers, traders, retailers, and service providers to demonstrate commitment to good environmental, social, ethical, and food safety practices.

- Over 400 standards worldwide.
- Build awareness and policy support, create clear guidelines and goals, have third-party verification, maintain consistency within a portfolio while developing a Green economy.

Few Sustainability Protocols

1. LEED (Leadership in Energy & Environmental Design)
2. WELL
3. Fitwel
4. Living Building Challenge
5. BREAM
6. Passive House
7. National Green Building Standard
8. Built Green
9. Evergreen Sustainable Development Standard (ESDS)

- Introduction of eco-labels and standards for organic food and other food products.
- Triple bottom line includes a set of practices or criteria for sustainably growing a crop or ethically harvesting a resource.
- Includes responsible fishing practices that do not endanger marine biodiversity and respect for human rights and fair wages.
- Sustainability protocols are accompanied by a verification process (certification) to evaluate enterprise compliance with a standard and traceability process for certified products.
- Focus on capacity building and working with stakeholders to support small, social, and environmental improvements.

Sustainable Development

Defined as meeting the needs of the present without compromising the ability of future generations to meet their own needs.

Aim of True Sustainable Development

- Optimum use of natural resources with a high degree of sustainability.
 - Minimum wastage.
 - Least generation of toxic by-products.
 - Maximum productivity.
1. **Inter-generational Equity:**
 - Hand over a safe, healthy, and resourceful environment to future generations.
 2. **Intra-generational Equity:**
 - Technological development of rich countries should support the economic growth of poor countries and narrow the wealth gap.
 3. Place more emphasis on pollution prevention and waste reduction.
 4. Recycle and reuse as many waste products and resources as possible.
 5. Make more goods that last longer and are easy to use, recycle, and repair.
 6. Depend on renewable sources of energy.
 7. Sustain Earth's biodiversity.
 8. Discourage Earth-degrading activities.
 9. Reduce poverty and rate of population growth.
 10. Don't use high-quality energy to do a job.

Goals of Sustainable Development

There are a total of 17 goals:

1. End poverty everywhere.
2. End hunger.
3. Ensure good health and promote well-being.
4. Ensure inclusive and equitable quality education.
5. Achieve gender equality and empower all women and children.
6. Ensure the availability and sustainable management of water and sanitation for all.
7. Ensure access to affordable, reliable, clean, and modern energy for all.
8. Promote sustainable economic growth.
9. Build resilient infrastructure and promote sustainable industrialization.
10. Reduce inequality within and among countries.
11. Make cities and human settlements inclusive.
12. Ensure sustainable consumption and production.
13. Take urgent action to combat climate change.
14. Conserve and sustainably use the oceans.
15. Protect and restore terrestrial ecosystems.
16. Promote peaceful and inclusive societies.

17. Strengthen the means of implementation and revitalize the global partnership for sustainable development.

Sustainable Development Targets

1. Eradicate extreme poverty.
2. Reduce poverty by half.
3. Implement social protection systems.
4. Ensure equal rights to economic resources, natural resources and new technology and financial services.
5. Build resilience of the poor.
6. Ensure resource mobilization from developed to less developed countries.
7. End hunger and ensure access to food.
8. End all forms of malnutrition.
9. Double agricultural productivity and income of small-scale food producers.
10. Ensure sustainable food production systems.
11. Maintain genetic diversity of seeds and farmed animals.
12. Increase investment in rural infrastructure and agricultural research.
13. Correct trade restrictions in agricultural markets.
14. Ensure proper functioning of food commodity markets.
15. Reduce the global maternal mortality ratio to less than 70 per 100,000 live births.
16. End preventable deaths of newborns and children under 5 years of age.
17. End epidemics of AIDS, tuberculosis, malaria, and water-borne diseases.
18. Reduce premature mortality from non-communicable diseases.
19. Strengthen prevention and treatment of substance abuse.
20. Halve the number of global deaths and injuries from road accidents.
21. Ensure universal access to sexual and reproductive healthcare services like family planning.
22. Achieve universal health coverage including financial risk protection.
23. Reduce deaths and illnesses from hazardous chemicals and pollution.

Sustainable Development Indicators

1. Proportion of the population living below the international poverty line.
2. Proportion of the population living below the national poverty line.
3. Proportion of population covered by social protection floors.
4. Proportion of adult population with secure tenure rights to land.
5. Proportion of local governments that adopt and implement local disaster risk reduction strategies.
6. Proportion of total government spending on essential services.
7. Prevalence of undernourishment.
8. Prevalence of stunting and malnutrition.

9. Average income of small-scale food producers.
10. Proportion of agricultural area under productive and sustainable practices.
11. Proportion of local breeds classified as being at risk of extinction.
12. Total official flows to the agricultural sector.
13. Agricultural export subsidies.
14. Indicator of food price anomalies.
15. Maternal mortality ratio.
16. Under-5 mortality rate.
17. Hepatitis B or Tuberculosis incidence per 100,000 population. Malaria incidence or HIV infections per 1000 population.
18. Suicide mortality rate.
19. Alcohol per capita consumption.
20. Death rate due to road traffic injuries.
21. Proportion of women of reproductive age who have their need for family planning.
22. Coverage of essential health services.
23. Mortality rate due to unsafe water, sanitation, and lack of hygiene.

Intervention Areas of Sustainable Development Goals

Implementation is difficult because of:

1. Climate change.
2. Use of natural resources.
3. Waste production.
4. Water pollution.
5. Deforestation.
6. Overfishing.
7. Poverty.
8. Ocean acidification.
9. Air pollution.

Climate Change

Refers to long-term shifts in temperatures and weather patterns; may be natural or artificial.

Causes of Climate Change

1. Presence of greenhouse gases increases global temperature.
2. Depletion of the ozone layer increases global temperature.
3. Uneven distribution of rainfall.
4. Rotation of Earth on its axis.
5. Seasonal changes.

Effects/Issues of Climate Change

1. Disturbs agriculture, leading to migration.
2. Upsetting the hydrological cycle, results in floods and droughts.
3. Disturbs global patterns of winds and ocean currents.
4. More frequent and intense droughts.
5. Sea levels rise.
6. Melting glaciers and warming oceans harm animals.
7. Higher temperature increases heat-related illnesses.
8. Wildfires start more easily.
9. Global surface temperature increases as greenhouse concentration increases.

Possible Solutions to Climate Change

1. Avoid burning fossil fuels.
2. Use renewable energy sources.
3. Reduce petrol and diesel vehicles.
4. Insulate and heat homes in a green way.
5. Reduce meat and dairy consumption.
6. Plant more trees.
7. Protect forests.
8. Oceans absorb CO₂, helping to stabilize climate.
9. Reduce overall consumption in wealthy countries.
10. Avoid using plastics.
11. Reduce carbon pollution.
12. Provide financial support to developing countries.

Case Studies

Climate Change on Chennai, East Coast Road (ECR) and Old Mahabalipuram Road (OMR)

- High climate impact due to Chennai's land use patterns, population stress and abuse of natural resources.
- Climate-induced impacts like drought, floods, heavy rains, and winds are becoming increasingly evident.
- Summer water crises and monsoonal flood disasters.
 - **Reason for Flood in Chennai:**
 - Changed the natural hydrology of the city like Land use Pattern.
 - Waterways are blocked by buildings and artificial structures.
 - **Remedy:**
 - Climate is unpredictable and aggressive.
 - Reduce emission of greenhouse gases.

Climate Change on Chennai, Ennore

- Ennore thermal power station (ETPS) violated air pollution norms for nearly 60%, responsible for drastic climate change.
- More than 56 lakh tonnes of coal ash is spread over the river bed.
 - **Issues/Effects:**
 1. Particulate matter pollution exceeded carrying capacity.
 2. SO₂ and NO₂ pollution exceeded carrying capacity.
 3. Children and women are affected, Gynecological problems reported.
 4. Natural drainage patterns altered due to fly ash pond construction.
 5. Rising sea levels, increased heat stress, intense rain events, droughts, and ocean desertification.
 6. Discharge of 120 million liters of hot wastewater daily will create localized marine deserts.

Carbon Credit

Definition

A tradable permit or certificate that represents the right to emit 1 ton of CO₂ or the equivalent amount of greenhouse gas.

Concept

- Kyoto Protocol introduced the concept.
- Aims to manage and reduce carbon dioxide emissions and greenhouse gases.
- One carbon credit is equal to one ton of carbon dioxide.
- Countries need to reduce their emissions by 5.2% compared to recorded numbers.
- Countries and companies need to reduce carbon emissions without buying credits.
- Less purchase means less carbon release into the atmosphere.

Types of Carbon Credits

1. Voluntary Emissions Reduction (VER):

- Exchanged in the over-the-counter market.

2. Certified Emissions Reduction (CER):

- Created through a regulatory framework to offset emissions.

How to Get Carbon Credit

Carbon credits and carbon markets mitigate the growth in greenhouse gases.

- Equal to one ton of CO₂ or CO₂ equivalent gases.
- Companies sell carbon credits to lower carbon footprint.
- Buyers and sellers use an exchange platform to trade.

Advantages and Disadvantages of Carbon Credits

- **Advantages:**

1. Corresponds to one ton of carbon not emitted.
2. Advantage for companies without alternatives to reduce emissions.
3. Favorable for company image.
4. Shows care for sustainable activities.
5. Enables companies to support decarbonization.

- **Disadvantages:**

1. Institutions can continue emitting greenhouse gases through the exchange market.
2. Companies do not invest in action to avoid emissions.
3. Reduction of 1 ton of carbon will never be enough.
4. It is an emergency solution, not depend on others to save carbon.
5. Cannot stop the negative impacts of global warming.

Carbon Footprint

Definition

The total amount of greenhouse gases (including *CO₂* and *CH₄*) that are generated by direct and indirect activities.

Individual Carbon Footprint

The sum total of direct and indirect carbon emissions over a year.

- Smaller carbon footprint: better for the future.
- Bigger carbon footprint: bigger negative impact on the environment.
- Average carbon footprint for a person in the United States is 16 tons. Globally, the average is closer to 4 tons.
- To avoid a 2°C rise in global temperatures, the average global carbon footprint per year needs to drop under 2 tons by 2050.

Sources of Carbon Footprint

1. Climate change.
2. Natural processes like volcanos.
3. Greenhouse gases emitted from human activities.
4. Pollution released by humans.
5. Transportation accounted for about 28% of total country.
6. Electricity generation accounted for about 28%.
7. Industrial activities 22%.
8. Heating and cooling in homes and businesses contribute 11%.

Causes of a Carbon Footprint

1. Food (especially meat (beef)).
2. Consumption.
3. Transportation.
4. Household energy.

How to Lower (Control) Carbon Footprint

Making small changes in our actions will reduce the carbon footprint.

1. Calculate your carbon footprint.
2. Drive less.
3. Switch to an electric or hybrid car.
4. Travel smart.
5. Switch to renewable energy.
6. Consider solar panels.
7. Make your home more efficient.
8. Turn your thermostat just 2 degrees cooler in winter and 2 degrees warmer in summer.
9. Get energy-efficient appliances.
10. Unplug electrical devices when not in use.
11. Buy locally-sourced food.
12. Start a home garden.
13. Eat less meat.
14. Don't waste water.
15. Reduce, reuse, and recycle.